

PACT Framework

Harmonia - The hybrid tabletop game of the Museo Internazionale e Biblioteca della Musica di Bologna

Authors

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PEOPLE

Who is the main audience?

Primary target: Young people / Students (15–25 years old)

Young people and students who live in Bologna and the surrounding areas. During the briefing with the museum, it emerged that despite free entry for under 18s and discounted tickets for university students, this age group is almost absent from the museum. They may encounter the museum as part of a school trip or independently, during their free time. The game can also be adopted in school settings as an alternative and engaging learning method, connecting classroom education with local cultural heritage. The game works as an entry point: it lowers the cultural barrier and turns a space often seen as "passive" or "boring" into something accessible and fun.

Secondary target: Adults with no background in classical music (26+)

Adults who live in Bologna and the surrounding areas, who have limited or no background in classical music but are open to cultural experiences, and who have never had a compelling reason to visit the museum. The game introduces key characters and stories from the collection, lowers the cultural barrier, and sparks curiosity, giving visitors some basic knowledge needed for a better understanding of the collection's items and context. TripAdvisor reviews show that visitors tend to enjoy the museum much more when they have some context and guidance, through the audio guide or a guided tour led by museum staff.

Main needs and characteristics?

Both groups need a gentle introduction to the museum's content and its historical context, one that is capable of triggering curiosity and motivating them to visit the museum, while also making the visit itself more enjoyable and meaningful, as players will already be familiar with the key characters, stories, and items of the collection before they even step inside. Both targets are comfortable with smartphones and digital interactions, but have limited knowledge of 18th-century music history and the figures in the collection.

More broadly, both target audiences share a need to feel genuinely engaged in the cultural experience, not passive observers. Several overlapping barriers prevent this engagement from occurring naturally:

The passivity of the existing Virtual Tour represents a critical gap in terms of embodied cognition and embodied interaction. The current experience is primarily spatial and architectural, users navigate rooms but cannot interact with the collection objects. This conflicts with what research on embodied experience tells us: cognition is not only mental but situated in the body. Involving the body through tangible interaction, physical manipulation, and agency over one's actions produces deeper learning, memorisation, and a stronger sense of presence.

People's attention threshold is particularly low when the interaction is passive and non-responsive. Being exposed daily to highly stimulating, interactive digital content, this audience finds the simple exploration of virtual rooms insufficiently engaging. The absence of narrative goals, competitive dynamics, or any sense of agency makes the Virtual Tour experience feel irrelevant to their habits and expectations.

Museum fatigue without targeted design strategies to capture and sustain attention, visitors rapidly lose focus. The game actively contributes to reducing museum fatigue by addressing one of its primary causes: **cognitive overload**. The overwhelming amount of information, and captions to read/listen during a museum visit rapidly saturates the visitor's attention capacity. By introducing key characters, stories, and objects of the collection before the visit, the game ensures that part of this cognitive load has already been processed in a playful and engaging context, making the museum experience lighter, more focused, and more enjoyable.

The hybrid game addresses all these barriers simultaneously: it introduces **agency** through competitive and cooperative mechanics, **embodied interaction** through tangible cards and physical manipulation, **narrative engagement** through scenario-based storytelling, and **attention management** through progressive gamification and rewarding structures.

ACTIVITIES

Scope and Goal: The Museo Internazionale e Biblioteca della Musica engaged our team as UX and Interaction Designers with the goal of expanding and strengthening the communicative and educational potential of the museum's existing Official Virtual Tour (VN 360° project), using it as a starting point for the design of a hybrid game. The game introduces key characters and stories from the collection, lowers the cultural barrier, and sparks curiosity, giving visitors some basic knowledge needed for a better understanding of the collection's items and context.

Temporal Aspects: The game is played in a single session of approximately 1–2 hours, in a casual setting such as home, a park, or a café. It is designed to be experienced before the museum visit, building basic knowledge and sparking curiosity, but it can also be played after the visit, giving continuity to the museum experience and reinforcing what the players have seen in person.

Cooperation: Competitive. Some missions are individual and competitive, others require team cooperation, but at the end only one person wins the game. The game supports 1 to 6 players.

Complexity: Moderate. The app guides players step by step, so no prior knowledge is required.

Security: No security concerns.

Content: The game is designed to be portable: the box contains a small fabric pouch that can hold all three card decks, making it easy to carry and play anywhere. The physical museum cards include QR codes linking to the Virtual Tour, in which we would like to implement 3D photogrammetry models of the actual collection objects, compensating for the limitations of the existing virtual tour, which does not allow users to explore items up close, zoom in, or view them at 360 degrees. The physical deck includes museum cards, malus cards, and theory cards. The app delivers narrative text, images of museum items and it manages the game flow.

CONTEXT

Physical Space: Informal domestic or outdoor settings such as a living room, park, or café. No museum infrastructure required. A flat surface is needed for the card deck. Each player needs a smartphone, and a connection.

Social Context: Played in small groups of friends or family (1–6 players) who know each other.

Organisational Context: The museum engaged our team as UX and Interaction Designers to expand the communicative and educational potential of its Official Virtual Tour (VN 360°), developed in collaboration with the Italian-Japanese communication studio Veronesi Namioka.

TECHNOLOGIES

Media: Registered sounds, narrative text, illustrated avatar images, reproductions of museum items, and 3D photogrammetry models accessible via QR code on the museum cards, visual tour. Scoring is managed through a digital ballot box inside the app.

Hardware: One smartphone per player (iOS/Android) to run the companion app, it scans QR codes and it gives the haptic feedback which is vibration. Physical card deck as the tangible game component.

Network: Online connection is required for accessing photogrammetry models and multiplayer synchronisation via Wi-Fi or Bluetooth local network.